Zhang cellulase and β-glucanase reproduction — benchmark report

Agent-alone candidates, published prediction values and reference variants calculated in this work.

Agent Scientist: gpt-6.1-sol. Predictor names and units are listed with each measurement.

Published predictions and our predictions

Property / predictor	Published prediction	Our candidates
Cellulase WT: Pythia (unitless)	T131D −12.587	T131D −12.79070377
Cellulase M2: ESM (unitless)	M2 + T257K −8.872	M2 + T257K −8.87227321
β-glucanase WT: Pythia (unitless)	N65I −10.762	Same variant N65I −11.32239389; retained leader M30P −14.53135109
β-glucanase M5: ESM (unitless)	M5 + I65Y −8.583; M5 + P10D −6.767	Same I65Y −2.72094822; retained leader P10D −6.70656031

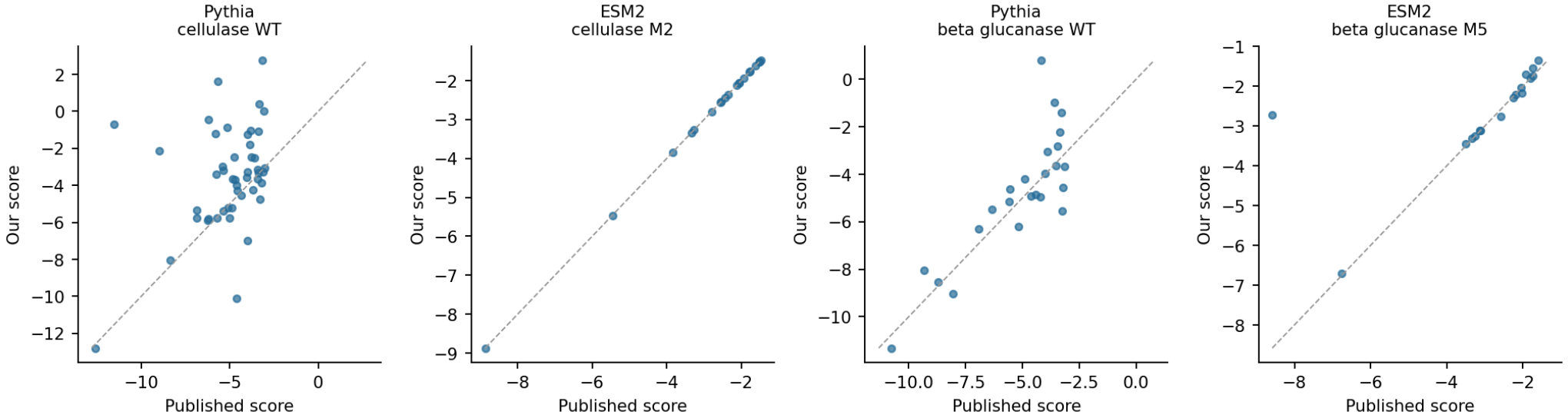
Source: Zhang et al., original supplementary tables S1/S3/S5/S7

Same-variant prediction comparison

Background	Predictor	Mutation	Published score	Our score
cellulase_WT	Pythia	T131D	-12.587	-12.790703773498535
cellulase_M2	ESM2	T257K	-8.872	-8.872273206710815
beta_glucanase_WT	Pythia	N65I	-10.762	-11.322393894195557
beta_glucanase_M5	ESM2	I65Y	-8.583	-2.7209482192993164
beta_glucanase_M5	ESM2	P10D	-6.767	-6.70656031370163

Predictor / background	Rows	Mean absolute difference	Rank correlation
Pythia / cellulase_WT	47	2.030403	0.310130
ESM2 / cellulase_M2	20	0.000221	0.999624
Pythia / beta_glucanase_WT	24	1.059948	0.744348
ESM2 / beta_glucanase_M5	17	0.417425	0.917229

Published versus recalculated scores

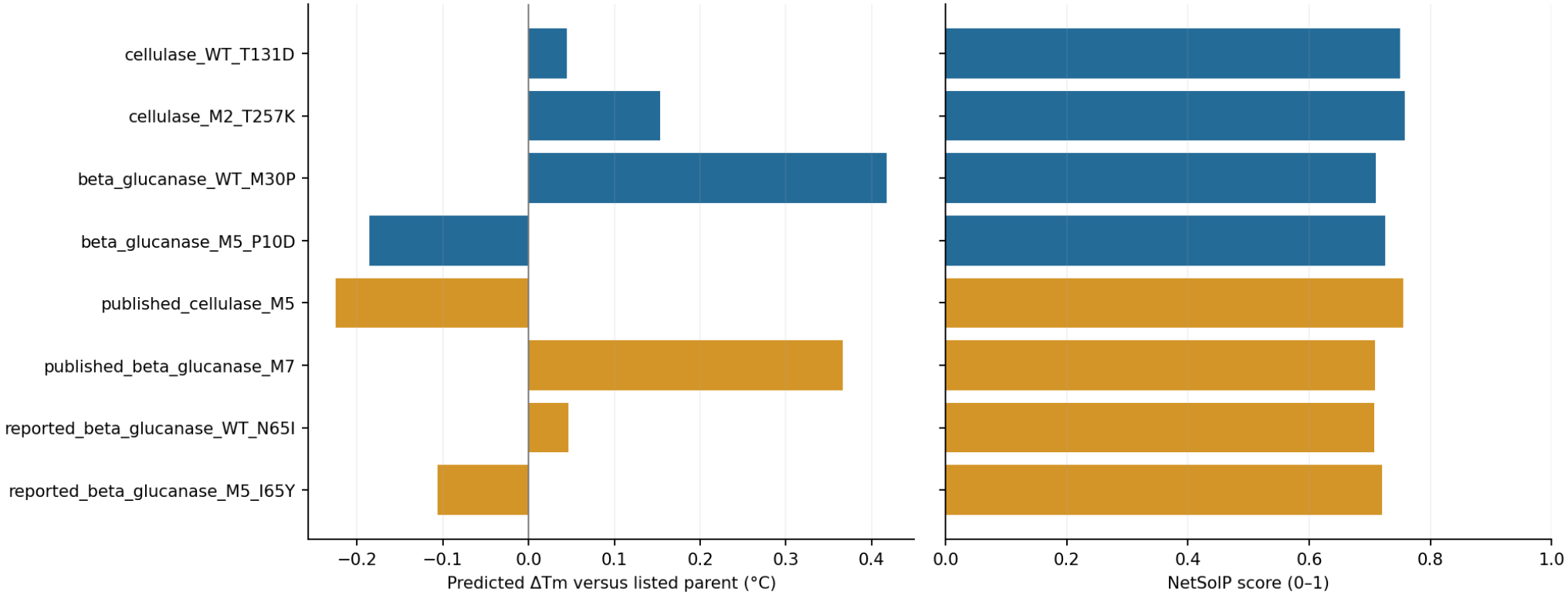


Each panel compares the same variants and scoring background. Scores are unitless; the dashed line marks equal values.

Property predictions calculated in this work

Sequence	Group	Parent / background	Predicted Tm (°C)	ΔTm (°C)	NetSolP	pHPred (raw)
Zhang_cellulase_WT_T131D	Our candidate	Zhang_cellulase_WT	47.2853	0.0444	0.750243	7.3643
Zhang_cellulase_M2_T257K	Our candidate	Zhang_cellulase_M2	47.4771	0.1533	0.757610	7.3643
Zhang_beta_glucanase_WT_M30P	Our candidate	Zhang_beta_glucanase_WT	48.1294	0.4176	0.710361	7.1286
Zhang_beta_glucanase_M5_P10D	Our candidate	Zhang_beta_glucanase_M5	47.1254	-0.1853	0.725672	7.1286
Zhang_published_cellulase_M5	Reference	Zhang_cellulase_WT	47.0161	-0.2248	0.755299	7.3643
Zhang_published_beta_glucanase_M7	Reference	Zhang_beta_glucanase_WT	48.0782	0.3664	0.708764	7.1286
Zhang_reported_beta_glucanase_WT_N65I	Reference	Zhang_beta_glucanase_WT	47.7586	0.0468	0.707348	7.1286
Zhang_reported_beta_glucanase_M5_I65Y	Reference	Zhang_beta_glucanase_M5	47.2043	-0.1065	0.719984	7.1143

Candidate and reference property predictions



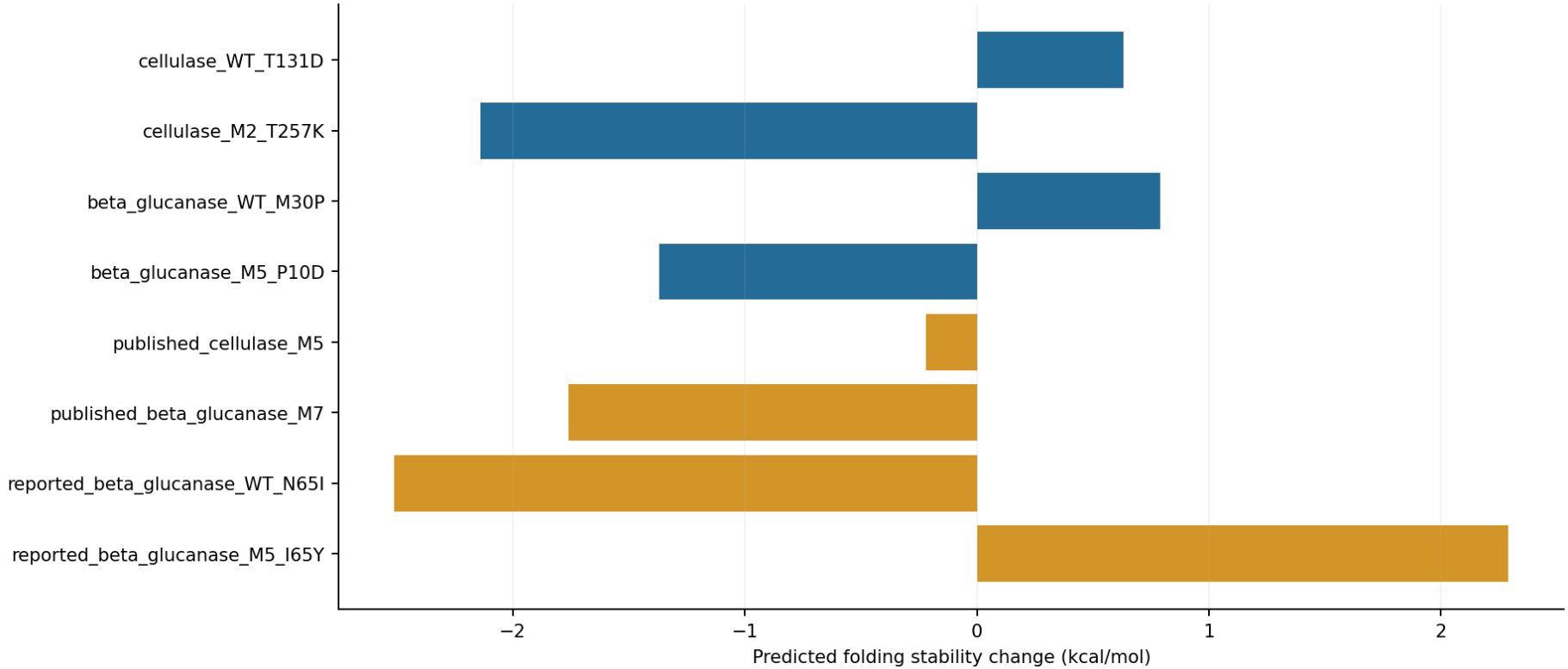
Blue: our candidates. Amber: reference variants calculated in this work. Changes use the parents listed in the table.

Predicted folding stability change — open-source assessment

Method: EvoEF1 1.1. Units: kcal/mol. Protocol: common WT; protein-only; 3 repair runs; 10 matched WT/mutant repacking runs; negative predicts stabilization. Values below were calculated in this work for the listed sequences.

Sequence	Group	Parent / background	Prediction (kcal/mol)
Zhang_cellulase_WT_T131D	Our candidate	Zhang_cellulase_WT	0.63
Zhang_cellulase_M2_T257K	Our candidate	Zhang_cellulase_WT	-2.14
Zhang_beta_glucanase_WT_M30P	Our candidate	Zhang_beta_glucanase_WT	0.79
Zhang_beta_glucanase_M5_P10D	Our candidate	Zhang_beta_glucanase_WT	-1.37
Zhang_published_cellulase_M5	Reported reference	Zhang_cellulase_WT	-0.22
Zhang_published_beta_glucanase_M7	Reported reference	Zhang_beta_glucanase_WT	-1.76
Zhang_reported_beta_glucanase_WT_N65I	Reported reference	Zhang_beta_glucanase_WT	-2.51
Zhang_reported_beta_glucanase_M5_I65Y	Reported reference	Zhang_beta_glucanase_WT	2.29

Predicted folding stability change: reference and our sequences



Amber: reported reference variants recalculated here. Blue: our retained candidates. All rows in this panel use the stated method and protocol.

Parent identity, preparation details and run records: results/task_006_zhang_hydrolases/agent-alone/README.md.